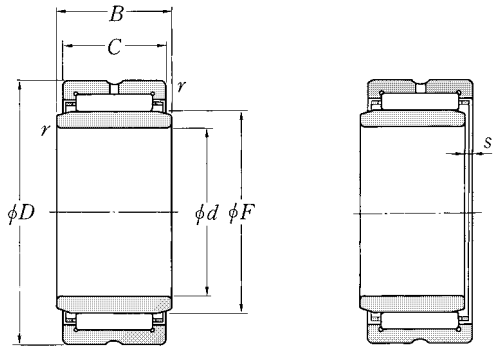



| Limiting speeds r/min grease | Limiting speeds r/min oil | Bearing numbers | Abutment dimensions mm $d_a$ min | Abutment dimensions mm $D_a$ max | Abutment dimensions mm $r_{as}$ max | Mass kg (approx.) |
|---|---|---|---|---|---|---|
| 6 000 | 9 000 | **MR283716+MH-232816** | 43 | 52 | 1.5 | 0.264 |
| 6 000 | 9 000 | **MR283720+MH-232820** | 43 | 52 | 1.5 | 0.330 |
| 6 000 | 9 000 | **MR283716+MH-242816** | 43 | 52 | 1.5 | 0.246 |
| 6 000 | 9 000 | **MR283720+MH-242820** | 43 | 52 | 1.5 | 0.307 |
| 5 500 | 8 000 | **MR324120+MH-243220** | 44.5 | 58.5 | 1.5 | 0.448 |
| 5 500 | 8 500 | **MR303920+MH-253020** | 46.5 | 55.5 | 1.5 | 0.368 |
| 5 500 | 8 000 | **MR324120+MH-253220** | 46 | 58.5 | 1.5 | 0.424 |
| 5 500 | 8 000 | **MR324116+MH-263216** | 48 | 58.5 | 1.5 | 0.317 |
| 5 500 | 8 000 | **MR324120+MH-263220** | 48 | 58.5 | 1.5 | 0.402 |
| 5 500 | 8 000 | **MR324116+MH-273216** | 49.5 | 58.5 | 1.5 | 0.300 |
| 5 500 | 8 000 | **MR324120+MH-273220** | 49.5 | 58.5 | 1.5 | 0.373 |
| 4 700 | 7 000 | **MR364824+MH-283624** | 51 | 69.5 | 1.5 | 0.732 |
| 4 700 | 7 000 | **MR364828+MH-283628** | 51 | 69.5 | 1.5 | 0.853 |
| 4 300 | 6 500 | **MR405224+MH-314024** | 56 | 74.5 | 2 | 0.848 |
| 4 300 | 6 500 | **MR405228+MH-314028** | 56 | 74.5 | 2 | 0.975 |
| 4 300 | 6 500 | **MR405224+MH-324024** | 59 | 74.5 | 2 | 0.812 |
| 4 300 | 6 500 | **MR405228+MH-324028** | 59 | 74.5 | 2 | 0.923 |
| 3 700 | 5 500 | **MR445628+MH-354428** | 63.5 | 81 | 2 | 1.07 |
| 3 700 | 5 500 | **MR445616+MH-364416** | 65 | 81 | 2 | 0.594 |
| 3 700 | 5 500 | **MR445624+MH-364424** | 65 | 81 | 2 | 0.884 |
| 3 700 | 5 500 | **MR445628+MH-364428** | 65 | 81 | 2 | 1.02 |
| 3 300 | 5 000 | **MR486028+MH-384828** | 68 | 87 | 2 | 1.25 |
| 3 300 | 5 000 | **MR486024+MH-404824** | 71.5 | 87 | 2 | 0.978 |
| 3 300 | 5 000 | **MR486028+MH-404828** | 71.5 | 87 | 2 | 1.14 |
| 3 300 | 5 000 | **MR526828+MH-425228** | 74.5 | 100 | 2 | 1.67 |
| 3 300 | 5 000 | **MR526832+MH-425232** | 74.5 | 100 | 2 | 1.92 |

## Inch series
## With inner ring

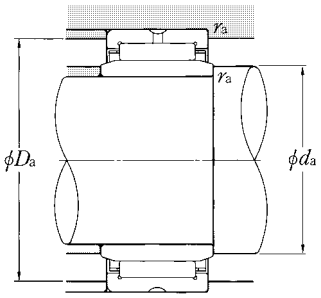
### Type MR＋MH




*d* 69.850～127.000mm

| Boundary dimensions mm (1/25.4mm) | | | | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf |
|---|---|---|---|---|---|---|---|---|---|---|
| *d* | *D* | *B* | *C* | $r_{s\,min}$ 1) | *F* | *s* 2) | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ |
| **69.850(2 3/4)** | 107.950(4 1/4) | 44.70 | 44.45(1 3/4) | 2 | 82.550 | 1.5 | 134 000 | 240 000 | 13 600 | 24 500 |
| | 107.950(4 1/4) | 51.05 | 50.80(2) | 2 | 82.550 | 3 | 146 000 | 268 000 | 14 900 | 27 400 |
| **74.612(2 15/16)** | 114.300(4 1/2) | 51.05 | 50.80(2) | 2 | 88.900 | 3 | 154 000 | 295 000 | 15 700 | 30 000 |
| **76.200(3)** | 114.300(4 1/2) | 44.70 | 44.45(1 3/4) | 2 | 88.900 | 3 | 141 000 | 264 000 | 14 400 | 26 900 |
| | 114.300(4 1/2) | 51.05 | 50.80(2) | 2 | 88.900 | 5.5 | 154 000 | 295 000 | 15 700 | 30 000 |
| **79.375(3 1/8)** | 120.650(4 3/4) | 51.05 | 50.80(2) | 2.5 | 95.250 | 5.5 | 162 000 | 320 000 | 16 500 | 32 500 |
| **82.550(3 1/4)** | 120.650(4 3/4) | 51.05 | 50.80(2) | 2.5 | 95.250 | 2.5 | 162 000 | 320 000 | 16 500 | 32 500 |
| | 127.000(5) | 51.05 | 50.80(2) | 2.5 | 101.600 | 2.5 | 169 000 | 345 000 | 17 200 | 35 500 |
| **85.725(3 3/8)** | 127.000(5) | 51.05 | 50.80(2) | 2.5 | 101.600 | 4.5 | 169 000 | 345 000 | 17 200 | 35 500 |
| **88.900(3 1/2)** | 127.000(5) | 51.05 | 50.80(2) | 2.5 | 101.600 | 2.5 | 169 000 | 345 000 | 17 200 | 35 500 |
| | 133.350(5 1/4) | 51.05 | 50.80(2) | 2.5 | 107.950 | 2.5 | 172 000 | 360 000 | 17 500 | 37 000 |
| **92.075(3 5/8)** | 133.350(5 1/4) | 51.05 | 50.80(2) | 2.5 | 107.950 | 4.5 | 172 000 | 360 000 | 17 500 | 37 000 |
| **95.250(3 3/4)** | 133.350(5 1/4) | 51.05 | 50.80(2) | 2.5 | 107.950 | 2.5 | 172 000 | 360 000 | 17 500 | 37 000 |
| | 152.400(6) | 57.40 | 57.15(2 1/4) | 2.5 | 114.300 | 3 | 238 000 | 435 000 | 24 300 | 44 500 |
| | 152.400(6) | 63.88 | 63.50(2 1/2) | 2.5 | 114.300 | 4 | 260 000 | 485 000 | 26 500 | 49 500 |
| **98.425(3 7/8)** | 152.400(6) | 57.40 | 57.15(2 1/4) | 2.5 | 114.300 | 5.5 | 238 000 | 435 000 | 24 300 | 44 500 |
| | 152.400(6) | 63.88 | 63.50(2 1/2) | 2.5 | 114.300 | 7 | 260 000 | 485 000 | 26 500 | 49 500 |
| **101.600(4)** | 165.100(6 1/2) | 51.05 | 50.80(2) | 2.5 | 127.000 | 2.5 | 227 000 | 425 000 | 23 200 | 43 000 |
| | 165.100(6 1/2) | 57.40 | 57.15(2 1/4) | 2.5 | 127.000 | 5.5 | 250 000 | 480 000 | 25 500 | 49 000 |
| | 165.100(6 1/2) | 63.88 | 57.15(2 1/4) | 2.5 | 127.000 | 7 | 273 000 | 535 000 | 27 800 | 54 500 |
| **107.950(4 1/4)** | 165.100(6 1/2) | 57.40 | 57.15(2 1/4) | 2.5 | 127.000 | 3 | 250 000 | 480 000 | 25 500 | 49 000 |
| **114.300(4 1/2)** | 177.800(7) | 63.88 | 63.50(2 1/2) | 2.5 | 139.700 | 5.5 | 285 000 | 585 000 | 29 100 | 59 500 |
| | 177.800(7) | 76.58 | 76.20(3) | 2.5 | 139.700 | 7 | 345 000 | 740 000 | 35 000 | 75 500 |
| **120.650(4 3/4)** | 184.150(7 1/4) | 82.55 | 76.20(3) | 3 | 146.050 | 3 | 360 000 | 775 000 | 36 500 | 79 000 |
| **127.000(5)** | 190.500(7 1/2) | 63.88 | 63.50(2 1/2) | 3 | 152.400 | 7 | 310 000 | 630 000 | 31 500 | 64 000 |
| | 190.500(7 1/2) | 76.58 | 76.20(3) | 3 | 152.400 | 5.5 | 375 000 | 800 000 | 38 000 | 81 500 |

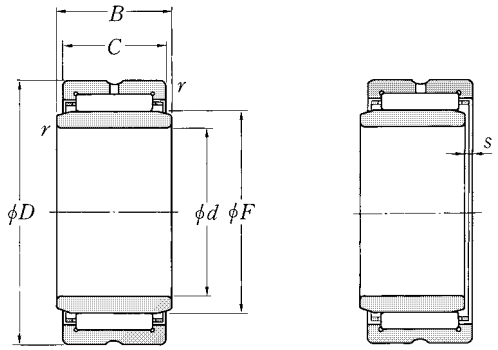
Note 1) Allowable minimum chamfer dimension $r$. 2) Allowable axial stroking value of inner ring against outer ring.




| Limiting speeds r/min grease | Limiting speeds r/min oil | Bearing numbers | Abutment dimensions mm $d_a$ min | Abutment dimensions mm $D_a$ max | Abutment dimensions mm $r_{as}$ max | Mass kg (approx.) |
|---|---|---|---|---|---|---|
| 3 300 | 5 000 | **MR526828 +MI-445228** | 78 | 100 | 2 | 1.55 |
| 3 300 | 5 000 | **MR526832 +MI-445232** | 78 | 100 | 2 | 1.77 |
| 3 000 | 4 500 | **MR567232 +MI-475632** | 83 | 106.5 | 2 | 2.00 |
| 3 000 | 4 500 | **MR567228 +MI-485628** | 84 | 106.5 | 2 | 1.65 |
| 3 000 | 4 500 | **MR567232 +MI-485632** | 84 | 106.5 | 2 | 1.92 |
| 2 800 | 4 200 | **MR607632 +MI-506032** | 88.5 | 111.5 | 2.5 | 2.15 |
| 2 800 | 4 200 | **MR607632 +MI-526032** | 91.5 | 111.5 | 2.5 | 1.99 |
| 2 600 | 3 900 | **MR648032 +MI-526432** | 91.5 | 118 | 2.5 | 2.39 |
| 2 600 | 3 900 | **MR648032 +MI-546432** | 94.5 | 118 | 2.5 | 2.22 |
| 2 600 | 3 900 | **MR648032 +MI-566432** | 98 | 118 | 2.5 | 2.05 |
| 2 500 | 3 700 | **MR688432 +MI-566832** | 98 | 124.5 | 2.5 | 2.53 |
| 2 500 | 3 700 | **MR688432 +MI-586832** | 101 | 124.5 | 2.5 | 2.36 |
| 2 500 | 3 700 | **MR688432 +MI-606832** | 104.5 | 124.5 | 2.5 | 2.17 |
| 2 300 | 3 500 | **MR729636 +MI-607236** | 104.5 | 143.5 | 2.5 | 4.02 |
| 2 300 | 3 500 | **MR729640 +MI-607240** | 104.5 | 143.5 | 2.5 | 4.47 |
| 2 300 | 3 500 | **MR729636 +MI-627236** | 107.5 | 143.5 | 2.5 | 3.77 |
| 2 300 | 3 500 | **MR729640 +MI-627240** | 107.5 | 143.5 | 2.5 | 4.19 |
| 2 100 | 3 100 | **MR8010432+MI-648032** | 110.5 | 156 | 2.5 | 3.60 |
| 2 100 | 3 100 | **MR8010436+MI-648036** | 110.5 | 156 | 2.5 | 4.92 |
| 2 100 | 3 100 | **MR8010440+MI-648040** | 110.5 | 156 | 2.5 | 5.47 |
| 2 100 | 3 100 | **MR8010436+MI-688036** | 117 | 156 | 2.5 | 4.44 |
| 1 900 | 2 900 | **MR8811240+MI-728840** | 123.5 | 169 | 2.5 | 6.04 |
| 1 900 | 2 900 | **MR8811248+MI-728848** | 123.5 | 169 | 2.5 | 7.26 |
| 1 800 | 2 700 | **MR9211648+MI-769248** | 132 | 172 | 3 | 7.48 |
| 1 700 | 2 600 | **MR9612040+MI-809640** | 140 | 177.5 | 3 | 6.54 |
| 1 700 | 2 600 | **MR9612048+MI-809648** | 140 | 177.5 | 3 | 7.84 |

## Inch series
## With inner ring

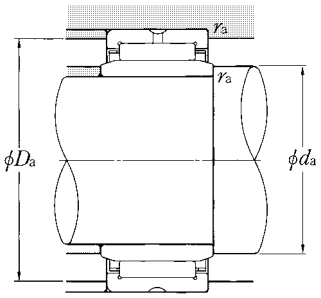
### Type MR＋MH




*d* 139.700～203.200mm

| Boundary dimensions mm (1/25.4mm) | | | | | | | Basic load ratings dynamic static N | | dynamic static kgf | |
|---|---|---|---|---|---|---|---|---|---|---|
| *d* | *D* | *B* | *C* | $r_{s\,min}$ 1) | *F* | *s* 2) | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ |
| **139.700(5 1/2)** | 203.200( 8) | 63.88 | 63.50(2 1/2) | 3 | 165.100 | 6 | 325 000 | 680 000 | 33 000 | 69 500 |
| | 203.200( 8) | 76.58 | 76.20(3) | 3 | 165.100 | 5.5 | 390 000 | 870 000 | 39 500 | 88 500 |
| **152.400(6)** | 231.775( 9 1/8) | 76.58 | 76.20(3) | 3 | 184.150 | 8.5 | 435 000 | 915 000 | 44 500 | 93 000 |
| **165.100(6 1/2)** | 244.475( 9 5/8) | 76.58 | 76.20(3) | 3 | 196.850 | 8.5 | 455 000 | 990 000 | 46 500 | 101 000 |
| **177.800(7)** | 257.175(10 1/8) | 76.58 | 76.20(3) | 3 | 209.550 | 8.5 | 475 000 | 1 060 000 | 48 500 | 109 000 |
| **190.500(7 1/2)** | 269.875(10 5/8) | 76.58 | 76.20(3) | 4 | 222.250 | 7 | 495 000 | 1 140 000 | 50 500 | 116 000 |
| **203.200(8)** | 282.575(11 1/8) | 76.58 | 76.20(3) | 4 | 234.950 | 7 | 510 000 | 1 210 000 | 52 000 | 124 000 |

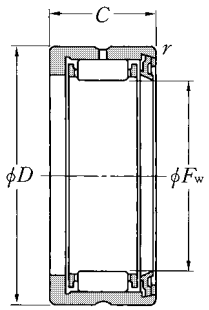
Note 1) Allowable minimum chamfer dimension $r$. 2) Allowable axial stroking value of inner ring against outer ring.



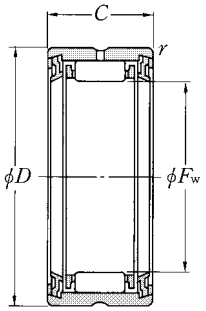

| Limiting speeds r/min grease | Limiting speeds r/min oil | Bearing numbers | Abutment dimensions mm $d_a$ min | Abutment dimensions mm $D_a$ max | Abutment dimensions mm $r_{as}$ max | Mass kg (approx.) |
|---|---|---|---|---|---|---|
| 1 600 | 2 400 | **MR10412840+MI-8810440** | 152.5 | 190 | 3 | 6.86 |
| 1 600 | 2 400 | **MR10412848+MI-8810448** | 152.5 | 190 | 3 | 8.22 |
| 1 500 | 2 200 | **MR11614648+MI-9611648** | 165.5 | 219 | 3 | 12.2 |
| 1 300 | 2 000 | **MR12415448+MI-10412448** | 178 | 231.5 | 3 | 13.1 |
| 1 300 | 1 900 | **MR13216248+MI-11213248** | 191 | 244 | 3 | 14.0 |
| 1 200 | 1 800 | **MR14017048+MI-12014048** | 206.5 | 254 | 4 | 15.0 |
| 1 100 | 1 700 | **MR14817848+MI-12814848** | 219 | 266.5 | 4 | 15.8 |

## Without inner ring

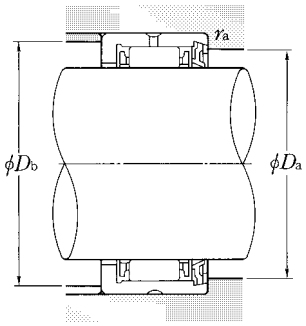
**Type RNA49··L**
**Type RNA49··LL**




**Type RNA49··L**
**(With single seal)**




**Type RNA49··LL**
**(With double seal)**



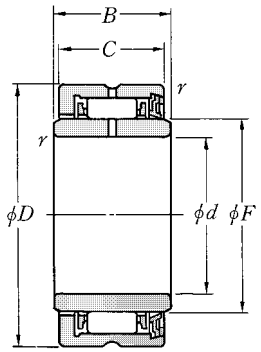

$F_w$ 14～58mm

| Boundary dimensions mm | | | | Basic load ratings N dynamic | static | kgf dynamic | static | Limiting speeds r/min grease | Bearing numbers Type RNA49··L | Type RNA49··LL | Abutment dimensions mm | | | Mass kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $D$ | $C$ | $r_{s\,min}$ [1] | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | $D_a$ max | $D_b$ max | $r_{as}$ max | (approx.) |
| **14** $^{+0.027}_{+0.016}$ | 22 | 13 | 0.3 | 7 200 | 8 500 | 735 | 865 | 10 000 | **RNA4900L** | **RNA4900LL** | 16 | 20 | 0.3 | 0.016 |
| **16** $^{+0.027}_{+0.016}$ | 24 | 13 | 0.3 | 7 750 | 9 700 | 795 | 990 | 10 000 | **RNA4901L** | **RNA4901LL** | 18 | 22 | 0.3 | 0.018 |
| **20** $^{+0.033}_{+0.020}$ | 28 | 13 | 0.3 | 8 300 | 11 200 | 845 | 1 150 | 10 000 | **RNA4902L** | **RNA4902LL** | 22 | 26 | 0.3 | 0.022 |
| **22** $^{+0.033}_{+0.020}$ | 30 | 13 | 0.3 | 8 500 | 11 900 | 865 | 1 220 | 9 000 | **RNA4903L** | **RNA4903LL** | 24 | 28 | 0.3 | 0.022 |
| **25** $^{+0.033}_{+0.020}$ | 37 | 17 | 0.3 | 15 200 | 19 900 | 1 550 | 2 030 | 8 000 | **RNA4904L** | **RNA4904LL** | 28 | 35 | 0.3 | 0.055 |
| **30** $^{+0.033}_{+0.020}$ | 42 | 17 | 0.3 | 16 000 | 22 600 | 1 640 | 2 300 | 6 500 | **RNA4905L** | **RNA4905LL** | 33 | 40 | 0.3 | 0.063 |
| **35** $^{+0.041}_{+0.025}$ | 47 | 17 | 0.3 | 18 000 | 27 400 | 1 830 | 2 800 | 5 500 | **RNA4906L** | **RNA4906LL** | 38 | 45 | 0.3 | 0.072 |
| **42** $^{+0.041}_{+0.025}$ | 55 | 20 | 0.6 | 22 700 | 39 500 | 2 320 | 4 000 | 4 800 | **RNA4907L** | **RNA4907LL** | 45 | 51 | 0.6 | 0.113 |
| **48** $^{+0.041}_{+0.025}$ | 62 | 22 | 0.6 | 27 800 | 53 500 | 2 830 | 5 450 | 4 200 | **RNA4908L** | **RNA4908LL** | 51 | 58 | 0.6 | 0.154 |
| **52** $^{+0.049}_{+0.030}$ | 68 | 22 | 0.6 | 28 600 | 57 000 | 2 920 | 5 800 | 3 800 | **RNA4909L** | **RNA4909LL** | 55 | 64 | 0.6 | 0.157 |
| **58** $^{+0.049}_{+0.030}$ | 72 | 22 | 0.6 | 30 500 | 64 000 | 3 100 | 6 500 | 3 400 | **RNA4910L** | **RNA4910LL** | 61 | 68 | 0.6 | 0.160 |

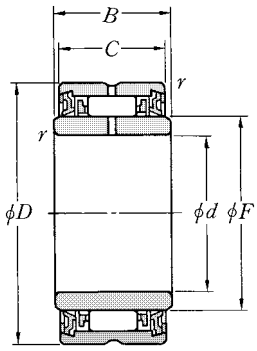
Note 1) Allowable minimum chamfer dimension $r$.

## With inner ring

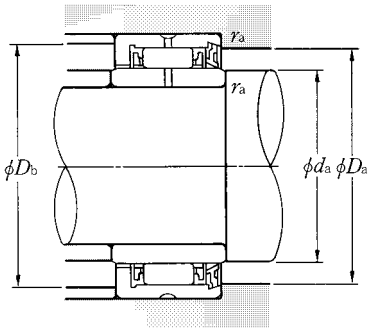
**Type NA49··L**
**Type NA49··LL**




**Type NA49··L**
**(With single seal)**




**Type NA49··LL**
**(With double seal)**

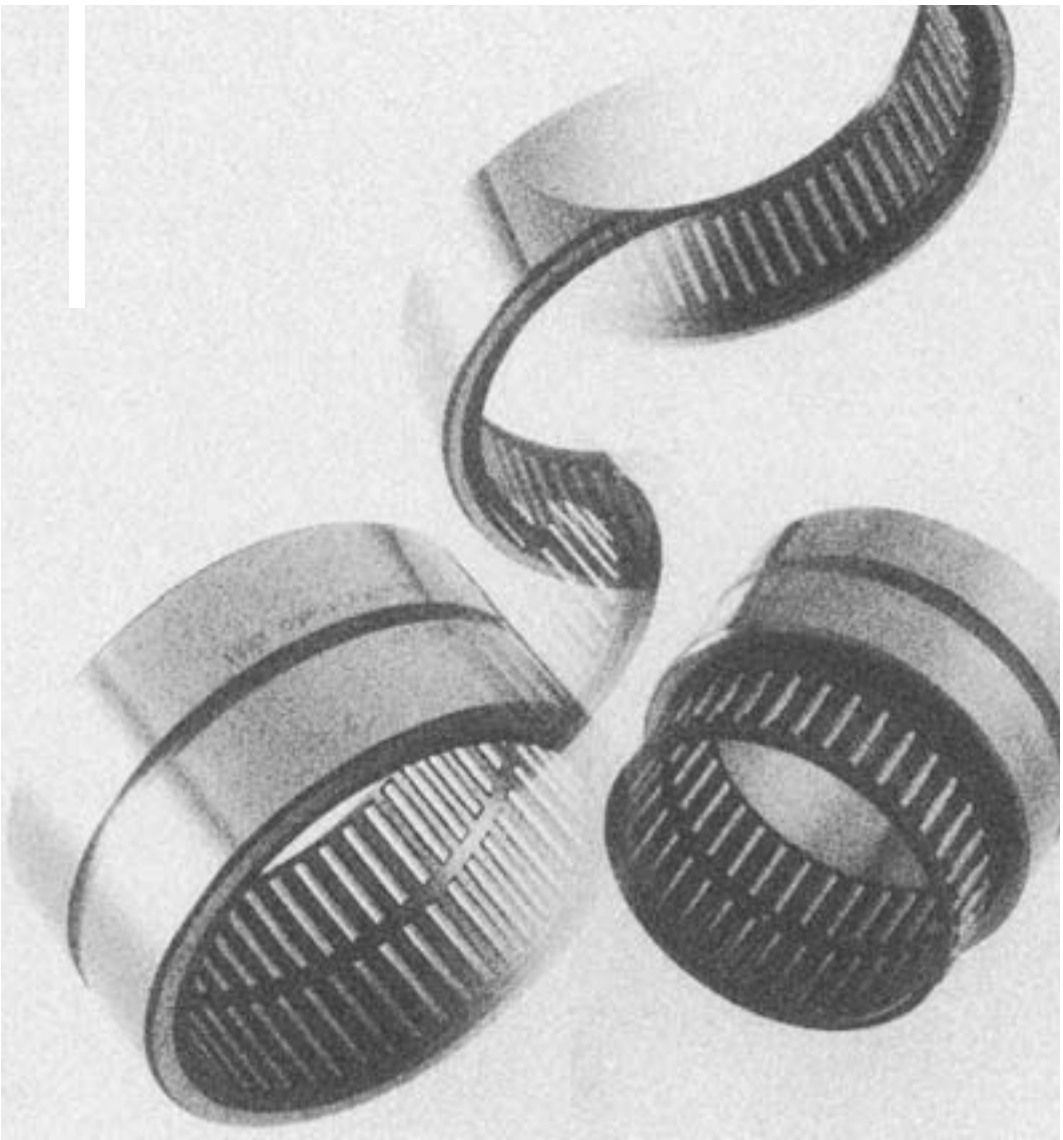



*d* 10～50mm

| Boundary dimensions mm | | | | | | Basic load ratings N dynamic | Basic load ratings N static | Basic load ratings kgf dynamic | Basic load ratings kgf static | Limiting speeds r/min grease | Bearing numbers Type NA49··L | Bearing numbers Type RNA49··LL | Abutment dimensions mm | | | | Mass kg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *d* | *D* | *B* | *C* | $r_{s\,min}$ 1) | *F* | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | | $d_a$ min | $D_a$ max | $D_b$ max | $r_{as}$ max | (approx.) |
| **10** | 22 | 14 | 13 | 0.3 | 14 | 7 200 | 8 500 | 735 | 865 | 10 000 | **NA4900L** | **NA4900LL** | 12 | 16 | 20 | 0.3 | 0.025 |
| **12** | 24 | 14 | 13 | 0.3 | 16 | 7 750 | 9 700 | 795 | 990 | 10 000 | **NA4901L** | **NA4901LL** | 14 | 18 | 22 | 0.3 | 0.028 |
| **15** | 28 | 14 | 13 | 0.3 | 20 | 8 300 | 11 200 | 845 | 1 150 | 10 000 | **NA4902L** | **NA4902LL** | 17 | 22 | 26 | 0.3 | 0.036 |
| **17** | 30 | 14 | 13 | 0.3 | 22 | 8 500 | 11 900 | 865 | 1 220 | 9 000 | **NA4903L** | **NA4903LL** | 19 | 24 | 28 | 0.3 | 0.039 |
| **20** | 37 | 18 | 17 | 0.3 | 25 | 15 200 | 19 900 | 1 550 | 2 030 | 8 000 | **NA4904L** | **NA4904LL** | 22 | 28 | 35 | 0.3 | 0.080 |
| **25** | 42 | 18 | 17 | 0.3 | 30 | 16 000 | 22 600 | 1 640 | 2 300 | 6 500 | **NA4905L** | **NA4905LL** | 27 | 33 | 40 | 0.3 | 0.093 |
| **30** | 47 | 18 | 17 | 0.3 | 35 | 18 000 | 27 400 | 1 830 | 2 800 | 5 500 | **NA4906L** | **NA4906LL** | 32 | 38 | 45 | 0.3 | 0.107 |
| **35** | 55 | 21 | 20 | 0.6 | 42 | 22 700 | 39 500 | 2 320 | 4 000 | 4 800 | **NA4907L** | **NA4907LL** | 39 | 45 | 51 | 0.6 | 0.175 |
| **40** | 62 | 23 | 22 | 0.6 | 48 | 27 800 | 53 500 | 2 830 | 5 450 | 4 200 | **NA4908L** | **NA4908LL** | 44 | 51 | 58 | 0.6 | 0.252 |
| **45** | 68 | 23 | 22 | 0.6 | 52 | 28 600 | 57 000 | 2 920 | 5 800 | 3 800 | **NA4909L** | **NA4909LL** | 49 | 55 | 64 | 0.6 | 0.290 |
| **50** | 72 | 23 | 22 | 0.6 | 58 | 30 500 | 64 000 | 3 100 | 6 500 | 3 400 | **NA4910L** | **NA4910LL** | 54 | 61 | 68 | 0.6 | 0.295 |

Note 1) Allowable minimum chamfer dimension $r$.

# Machined Ring Needle Roller Bearings, Separable Type

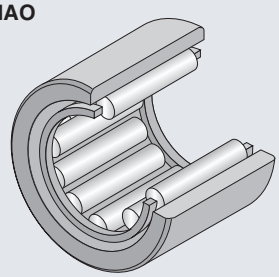
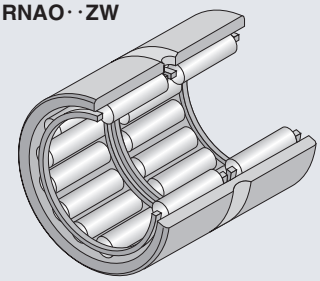
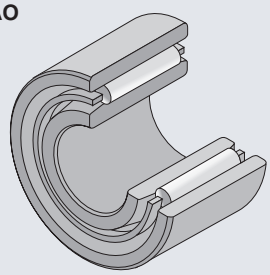
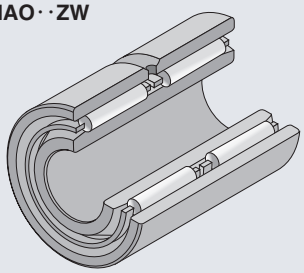


# Machined Ring Needle Roller Bearings, Separable Type

The machined ring of this bearing type has no rib or side plate and, hence, the outer ring and the needle roller and cage assembly are separable from each other.

The outer ring can't regulate axial displacement of the needle roller and cage assembly and, therefore, the bearing construction must be so designed that the needle roller and cage assembly can be guided by a shaft or a housing. Furthermore, the outer ring, the needle roller and cage assembly separable from one another, and the inner ring can be press-fitted individually in a shaft or a housing. This could facilitate the bearing mounting work.

This bearing type is suitable for an application requiring high running accuracy because the resultant radial clearance can be retained in a narrow range by selection and combination of appropriate inner ring, outer ring and needle roller and cage assembly.

| Bearing type | Applied shaft diameter (mm) | Composition of nominal bearing number | Remarks |
|---|---|---|---|
| **Type RNAO**<br>  | $\phi$5～$\phi$100 | **RNAO**－**10**×**17**×**10** **T2**<br>Tail code<br>Width<br>Outer diameter<br>Bore diameter<br>Type code<br>**[Tail code]**<br>T2 : Resin cage<br>ZW: Double-row type | Bearing with tail code T2 uses a polyamide resin cage and, therefore, it shall be used at allowable temperature 120˚C and, under continuous running, at 100˚C and less.<br><br>For an application requiring high running accuracy, manufacture of the bearings conforming to JIS Class-6, -5 and -4 is also available on special request. |
| **Type RNAO··ZW**<br>  | $\phi$8～$\phi$80 | | |
| **Type NAO**<br>  | $\phi$8～$\phi$90 | **NAO**－**10**×**17**×**10** **ZW**<br>Tail code<br>Width<br>Outer diameter<br>Bore diameter<br>Type code<br>**[Tail code]**<br>T2 : Resin cage<br>ZW: Double-row type | |
| **Type NAO··ZW**<br>  | $\phi$10～$\phi$70 | | |

## Accuracy of bearing

The dimensional accuracy, profile accuracy and running accuracy of machined ring needle roller bearing, separate type (with inner ring) are specified in JIS B 1514 (Accuracy of Rolling Bearings). (Refer to **Table 4.3** of Section 4. "**Bearing Tolerances**" on page A-26.) Although the accuracy of **NTN** standard bearings conforms to JIS Class-0, **NTN** can also supply bearings conforming to JIS Class-6, -5 and -4. Feel free to contact **NTN** for the further detail of these bearings.

The dimensional tolerances for the roller inscribed circle diameters ($F_w$) of the bearing type without inner ring conform to ISO Tolerance Range Class-F6. The outer ring and the needle roller and cage assembly are supplied in set and, therefore, the bearing must be installed with the combination of these two in set remained unchanged.

Particularly when high bearing accuracy is required, a bearing with grinding allowance for its inner ring raceway surface can be supplied upon request from a client. In that case, however, the client is requested to finish the inner ring to the intended dimension after having installed on a shaft.

## Radial clearance and bearing fits

**Table 5.1** of Subsection 5.1 "**Radial Clearance in Bearings**" (page A-30) shows the radial clearance of bearings with inner ring. Because of the narrow non-interchangeable clearance range, the bearings shipped after adjusted to a specific non-interchangeable clearance must be installed with the clearance remained unchanged.

The respective fitting tolerances for shaft and housing bore on/in which bearing with inner ring is press-fitted are per **Table 6.3** of 6.4 "**Recommended Bearing Fits**" (page A-34) which specifies them according to load characteristic, load magnitude, and shaft and housing bore sizes.

The profile accuracy and surface roughness of shaft and housing to be applied are as specified in **Table 7.3** of 7.3 "**Shaft and Housing Accuracy**" (page A-39).

A shaft is used as the direct raceway surface for a bearing without inner ring and, in this case, the shaft diameter (raceway diameter) tolerance is per Table 1 specifying the tolerances according to each running clearance. And K7 tolerance generally in broad use is applied as the dimensional tolerance for the housing bore. Feel free to contact **NTN** for application of housing bore tolerance other than K7.

For the profile accuracy, surface roughness and hardness of a shaft acting as the raceway surface, refer to **Table 7.4** of 7.4 "**Raceway Surface Accuracy**" (page A-39).

## Dimension of oil hole in outer ring

The outer ring of bearing Type ZW is provided with an oil hole and an oil groove to facilitate oil lubrication to bearing.

**Table 1** shows the oil hole dimension every outer ring diameter.

**Table 1 Oil hole dimension**

| Outer ring outer diameter | | Oil hole dia. (mm) | Number of oil hole |
|---|---|---|---|
| over | incl. | | |
| — | 20 | 2.0 | 1 |
| 20 | 40 | 2.5 | 1 |
| 40 | 80 | 3.0 | 1 |
| 80 | 200 | 3.5 | 1 |

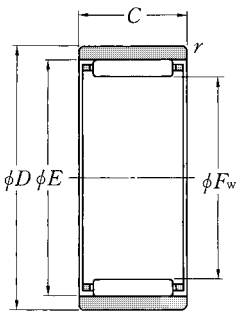
## Mounting relations

The inner ring and outer ring of any machined ring needle roller bearing, separable type must both be positioned in axial direction by shoulder or a snap ring.

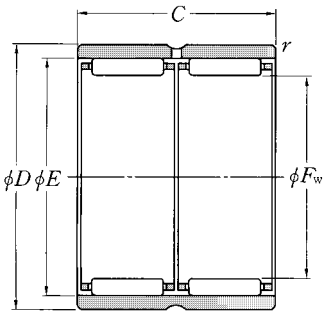
For the shoulder dimension and corner roundness ($r_a$) of the shaft and the housing are as specified in applicable "Dimensions Table". The cage must be guided by the shaft or the side face of the housing shoulder, but the guide surface must be finished by, at least, grinding for deburring.

## Without inner ring

### Type RNAO
### Type RNAO··ZW



Type RNAO

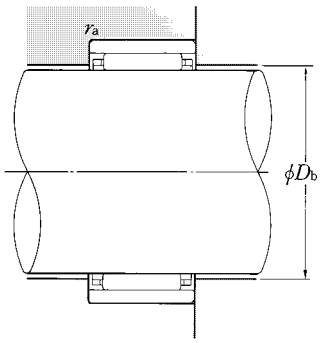


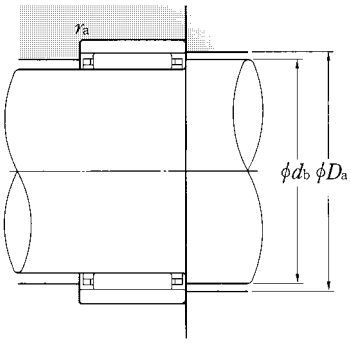
Type RNAO··ZW

$F_w$ 5～20mm

| Boundary dimensions mm | | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | oil | Bearing numbers |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $D$ | $C$ | $r_{s\,min}$ [1] | $E$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | |
| 5 +0.018 +0.010 | 10 | 8 | 0.15 | 8 | 2 640 | 2 190 | 269 | 224 | 27 000 | 40 000 | RNAO- 5×10×8T2 |
| 6 +0.018 +0.010 | 13 | 8 | 0.3 | 9 | 2 660 | 2 280 | 272 | 233 | 25 000 | 37 000 | RNAO- 6×13×8T2 |
| 7 +0.022 +0.013 | 14 | 8 | 0.3 | 10 | 2 670 | 2 350 | 272 | 239 | 23 000 | 34 000 | RNAO- 7×14×8T2 |
| 8 +0.022 +0.013 | 15 | 10 | 0.3 | 11 | 4 000 | 4 100 | 410 | 420 | 21 000 | 32 000 | RNAO- 8×15×10T2 |
| | 16 | 20 | 0.3 | 12 | 7 950 | 8 350 | 810 | 850 | 21 000 | 32 000 | RNAO- 8×16×20ZWT2 |
| 10 +0.022 +0.013 | 17 | 10 | 0.3 | 13 | 4 550 | 5 100 | 460 | 520 | 19 000 | 28 000 | RNAO-10×17×10T2 |
| | 20 | 12 | 0.3 | 16 | 7 100 | 5 950 | 720 | 610 | 19 000 | 28 000 | RNAO-10×20×12 |
| 12 +0.027 +0.016 | 19 | 13.5 | 0.3 | 15 | 6 000 | 7 700 | 615 | 785 | 17 000 | 26 000 | RNAO-12×19×13.5 |
| | 22 | 12 | 0.3 | 18 | 8 650 | 8 000 | 880 | 815 | 17 000 | 26 000 | RNAO-12×22×12 |
| 14 +0.027 +0.016 | 22 | 13 | 0.3 | 18 | 8 300 | 10 100 | 845 | 1 030 | 16 000 | 24 000 | RNAO-14×22×13 |
| | 22 | 20 | 0.3 | 18 | 11 800 | 16 000 | 1 210 | 1 630 | 16 000 | 24 000 | RNAO-14×22×20ZW |
| | 26 | 12 | 0.3 | 20 | 9 350 | 9 150 | 955 | 930 | 16 000 | 24 000 | RNAO-14×26×12 |
| 15 +0.027 +0.016 | 23 | 13 | 0.3 | 19 | 8 250 | 10 200 | 840 | 1 040 | 15 000 | 23 000 | RNAO-15×23×13 |
| | 23 | 20 | 0.3 | 19 | 11 700 | 16 100 | 1 200 | 1 640 | 15 000 | 23 000 | RNAO-15×23×20ZW |
| 16 +0.027 +0.016 | 24 | 13 | 0.3 | 20 | 9 050 | 11 800 | 925 | 1 200 | 15 000 | 23 000 | RNAO-16×24×13 |
| | 24 | 20 | 0.3 | 20 | 12 900 | 18 500 | 1 310 | 1 890 | 15 000 | 23 000 | RNAO-16×24×20ZW |
| | 28 | 12 | 0.3 | 22 | 11 700 | 12 500 | 1 190 | 1 280 | 15 000 | 23 000 | RNAO-16×28×12 |
| 17 +0.027 +0.016 | 25 | 13 | 0.3 | 21 | 9 400 | 12 600 | 960 | 1 280 | 15 000 | 22 000 | RNAO-17×25×13 |
| | 25 | 20 | 0.3 | 21 | 12 800 | 18 600 | 1 300 | 1 900 | 15 000 | 22 000 | RNAO-17×25×20ZW |
| | 25 | 26 | 0.3 | 21 | 16 100 | 25 200 | 1 640 | 2 570 | 15 000 | 22 000 | RNAO-17×25×26ZW |
| 18 +0.027 +0.016 | 26 | 13 | 0.3 | 22 | 8 900 | 11 900 | 910 | 1 210 | 14 000 | 21 000 | RNAO-18×26×13 |
| | 26 | 20 | 0.3 | 22 | 12 700 | 18 800 | 1 290 | 1 910 | 14 000 | 21 000 | RNAO-18×26×20ZW |
| | 30 | 12 | 0.3 | 24 | 12 300 | 13 800 | 1 250 | 1 410 | 14 000 | 21 000 | RNAO-18×30×12 |
| | 30 | 24 | 0.3 | 24 | 21 100 | 27 700 | 2 150 | 2 820 | 14 000 | 21 000 | RNAO-18×30×24ZW |
| 20 +0.033 +0.020 | 28 | 13 | 0.3 | 24 | 10 000 | 14 300 | 1 020 | 1 460 | 13 000 | 20 000 | RNAO-20×28×13 |
| | 28 | 26 | 0.3 | 24 | 17 100 | 28 600 | 1 750 | 2 910 | 13 000 | 20 000 | RNAO-20×28×26ZW |
| | 32 | 12 | 0.3 | 26 | 12 900 | 15 100 | 1 320 | 1 540 | 13 000 | 20 000 | RNAO-20×32×12 |

Note 1) Allowable minimum chamfer dimension $r$.

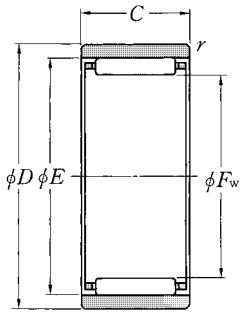


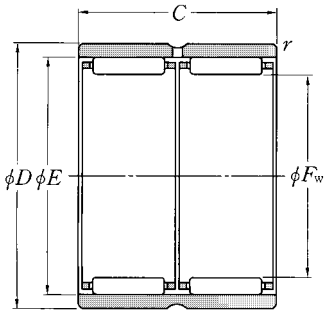

| Abutment dimensions mm | | | | Mass kg |
|---|---|---|---|---|
| $d_b$ | $D_a$ max | $D_b$ | $r_{as}$ max | (approx.) |
| 7.7 | 8.8 | 5.3 | 0.15 | 0.003 |
| 8.7 | 11 | 6.3 | 0.3 | 0.006 |
| 9.7 | 12 | 7.3 | 0.3 | 0.006 |
| 10.7 | 13 | 8.3 | 0.3 | 0.008 |
| 11.7 | 14 | 8.3 | 0.3 | 0.017 |
| 12.7 | 15 | 10.3 | 0.3 | 0.010 |
| 15.7 | 18 | 10.3 | 0.3 | 0.018 |
| 14.7 | 17 | 12.3 | 0.3 | 0.015 |
| 17.6 | 20 | 12.3 | 0.3 | 0.019 |
| 17.6 | 20 | 14.4 | 0.3 | 0.018 |
| 17.6 | 20 | 14.4 | 0.3 | 0.027 |
| 19.6 | 24 | 14.4 | 0.3 | 0.029 |
| 18.6 | 21 | 15.4 | 0.3 | 0.020 |
| 18.6 | 21 | 15.4 | 0.3 | 0.031 |
| 19.6 | 22 | 16.4 | 0.3 | 0.021 |
| 19.6 | 22 | 16.4 | 0.3 | 0.032 |
| 21.6 | 26 | 16.4 | 0.3 | 0.032 |
| 20.6 | 23 | 17.4 | 0.3 | 0.022 |
| 20.6 | 23 | 17.4 | 0.3 | 0.034 |
| 20.6 | 23 | 17.4 | 0.3 | 0.044 |
| 21.6 | 24 | 18.4 | 0.3 | 0.022 |
| 21.6 | 24 | 18.4 | 0.3 | 0.033 |
| 23.6 | 28 | 18.4 | 0.3 | 0.035 |
| 23.6 | 28 | 18.4 | 0.3 | 0.069 |
| 23.6 | 26 | 20.4 | 0.3 | 0.025 |
| 23.6 | 26 | 20.4 | 0.3 | 0.050 |
| 25.6 | 30 | 20.4 | 0.3 | 0.038 |

## Without inner ring

### Type RNAO
### Type RNAO··ZW




Type RNAO

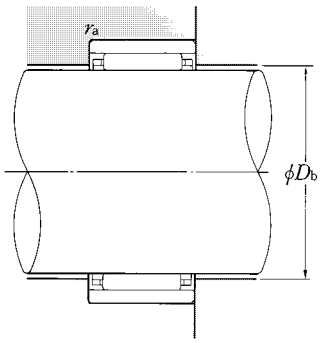


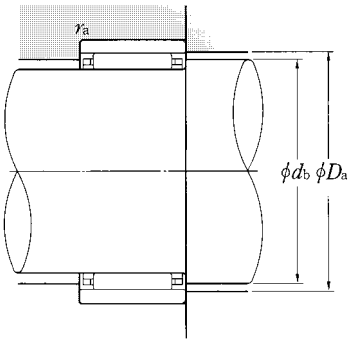

Type RNAO··ZW

$F_w$ 20～40mm

| Boundary dimensions mm | | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | oil | Bearing numbers |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | $D$ | $C$ | $r_{s\,min}$ 1) | $E$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | |
| **20** +0.033 +0.020 | 32 | 24 | 0.3 | 26 | 22 100 | 30 000 | 2 260 | 3 100 | 13 000 | 20 000 | **RNAO-20×32×24ZW** |
| | 30 | 13 | 0.3 | 26 | 10 200 | 15 200 | 1 040 | 1 550 | 12 000 | 18 000 | **RNAO-22×30×13** |
| **22** +0.033 +0.020 | 30 | 26 | 0.3 | 26 | 17 500 | 30 500 | 1 790 | 3 100 | 12 000 | 18 000 | **RNAO-22×30×26ZW** |
| | 35 | 16 | 0.3 | 29 | 18 700 | 22 700 | 1 910 | 2 310 | 12 000 | 18 000 | **RNAO-22×35×16** |
| | 35 | 32 | 0.3 | 29 | 32 000 | 45 500 | 3 300 | 4 650 | 12 000 | 18 000 | **RNAO-22×35×32ZW** |
| | 35 | 17 | 0.3 | 29 | 14 200 | 24 000 | 1 450 | 2 450 | 11 000 | 16 000 | **RNAO-25×35×17** |
| **25** +0.033 +0.020 | 35 | 26 | 0.3 | 29 | 18 400 | 33 500 | 1 880 | 3 450 | 11 000 | 16 000 | **RNAO-25×35×26ZW** |
| | 37 | 16 | 0.3 | 32 | 19 500 | 24 700 | 1 990 | 2 520 | 11 000 | 16 000 | **RNAO-25×37×16** |
| | 37 | 32 | 0.3 | 32 | 33 500 | 49 500 | 3 400 | 5 050 | 11 000 | 16 000 | **RNAO-25×37×32ZW** |
| **26** +0.033 +0.020 | 39 | 13 | 0.3 | 30 | 11 800 | 19 200 | 1 200 | 1 960 | 10 000 | 15 000 | **RNAO-26×39×13** |
| **28** +0.033 +0.020 | 40 | 16 | 0.3 | 35 | 21 200 | 28 400 | 2 160 | 2 900 | 9 500 | 14 000 | **RNAO-28×40×16** |
| | 40 | 32 | 0.3 | 35 | 36 500 | 57 000 | 3 700 | 5 800 | 9 500 | 14 000 | **RNAO-28×40×32ZW** |
| | 40 | 17 | 0.3 | 35 | 19 400 | 32 500 | 1 970 | 3 350 | 9 000 | 13 000 | **RNAO-30×40×17** |
| **30** +0.033 +0.020 | 40 | 26 | 0.3 | 35 | 25 200 | 46 000 | 2 570 | 4 650 | 9 000 | 13 000 | **RNAO-30×40×26ZW** |
| | 42 | 16 | 0.3 | 37 | 21 900 | 30 500 | 2 230 | 3 100 | 9 000 | 13 000 | **RNAO-30×42×16** |
| | 42 | 32 | 0.3 | 37 | 37 500 | 60 500 | 3 850 | 6 200 | 9 000 | 13 000 | **RNAO-30×42×32ZW** |
| **32** +0.041 +0.025 | 42 | 13 | 0.3 | 37 | 14 500 | 23 000 | 1 480 | 2 350 | 8 500 | 13 000 | **RNAO-32×42×13** |
| | 45 | 13 | 0.3 | 40 | 15 200 | 25 100 | 1 550 | 2 560 | 7 500 | 11 000 | **RNAO-35×45×13** |
| | 45 | 17 | 0.3 | 40 | 20 000 | 36 000 | 2 040 | 3 650 | 7 500 | 11 000 | **RNAO-35×45×17** |
| **35** +0.041 +0.025 | 45 | 26 | 0.3 | 40 | 26 100 | 50 000 | 2 660 | 5 100 | 7 500 | 11 000 | **RNAO-35×45×26ZW** |
| | 47 | 16 | 0.3 | 42 | 24 100 | 36 000 | 2 450 | 3 650 | 7 500 | 11 000 | **RNAO-35×47×16** |
| | 47 | 18 | 0.3 | 42 | 24 700 | 37 000 | 2 510 | 3 750 | 7 500 | 11 000 | **RNAO-35×47×18** |
| | 47 | 32 | 0.3 | 42 | 41 500 | 71 500 | 4 200 | 7 300 | 7 500 | 11 000 | **RNAO-35×47×32ZW** |
| **37** +0.041 +0.025 | 47 | 13 | 0.3 | 42 | 15 900 | 27 100 | 1 620 | 2 770 | 7 000 | 11 000 | **RNAO-37×47×13** |
| | 52 | 18 | 0.3 | 44 | 26 300 | 41 000 | 2 680 | 4 150 | 7 000 | 11 000 | **RNAO-37×52×18** |
| | 50 | 17 | 0.3 | 45 | 21 800 | 41 500 | 2 220 | 4 250 | 6 500 | 10 000 | **RNAO-40×50×17** |
| **40** +0.041 +0.025 | 50 | 34 | 0.3 | 45 | 37 500 | 83 000 | 3 800 | 8 500 | 6 500 | 10 000 | **RNAO-40×50×34ZW** |
| | 55 | 20 | 0.3 | 47 | 31 000 | 51 500 | 3 150 | 5 250 | 6 500 | 10 000 | **RNAO-40×55×20** |

Note 1) Allowable minimum chamfer dimension $r$.

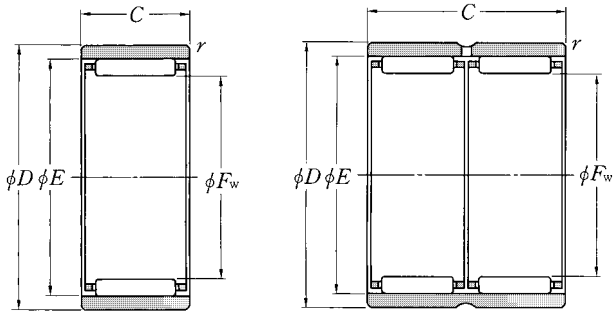



| Abutment dimensions mm | | | | Mass kg |
|---|---|---|---|---|
| $d_b$ | $D_a$ max | $D_b$ | $r_{as}$ max | (approx.) |
| 25.6 | 30 | 20.4 | 0.3 | 0.080 |
| 25.6 | 28 | 22.4 | 0.3 | 0.027 |
| 25.6 | 28 | 22.4 | 0.3 | 0.054 |
| 28.4 | 33 | 22.4 | 0.3 | 0.059 |
| 28.4 | 33 | 22.4 | 0.3 | 0.118 |
| 28.4 | 33 | 25.6 | 0.3 | 0.053 |
| 28.4 | 33 | 25.6 | 0.3 | 0.076 |
| 31.4 | 35 | 25.6 | 0.3 | 0.060 |
| 31.4 | 35 | 25.6 | 0.3 | 0.119 |
| 29.4 | 37 | 26.6 | 0.3 | 0.060 |
| 34.4 | 38 | 28.6 | 0.3 | 0.061 |
| 34.4 | 38 | 28.6 | 0.3 | 0.122 |
| 34.4 | 38 | 30.6 | 0.3 | 0.060 |
| 34.4 | 38 | 30.6 | 0.3 | 0.094 |
| 36.4 | 40 | 30.6 | 0.3 | 0.069 |
| 36.4 | 40 | 30.6 | 0.3 | 0.137 |
| 36.4 | 40 | 32.6 | 0.3 | 0.049 |
| 39.4 | 43 | 35.6 | 0.3 | 0.053 |
| 39.4 | 43 | 35.6 | 0.3 | 0.069 |
| 39.4 | 43 | 35.6 | 0.3 | 0.091 |
| 41.4 | 45 | 35.6 | 0.3 | 0.078 |
| 41.4 | 45 | 35.6 | 0.3 | 0.089 |
| 41.4 | 45 | 35.6 | 0.3 | 0.156 |
| 41.4 | 45 | 37.6 | 0.3 | 0.056 |
| 43.4 | 50 | 37.6 | 0.3 | 0.125 |
| 44.4 | 48 | 40.6 | 0.3 | 0.074 |
| 44.4 | 48 | 40.6 | 0.3 | 0.152 |
| 46.2 | 53 | 40.6 | 0.3 | 0.145 |

## Without inner ring

### Type RNAO
### Type RNAO··ZW

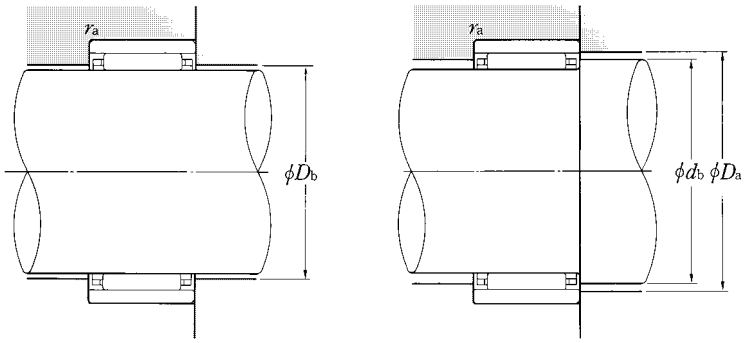



Type RNAO　　Type RNAO··ZW

$F_w$ 40～85mm

| Boundary dimensions mm | | | | | | Basic load ratings N dynamic | | kgf static | | Limiting speeds r/min grease | oil | Bearing numbers |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | | $D$ | $C$ | $r_{s\,min}$ [1] | $E$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | |
| **40** | +0.041 +0.025 | 55 | 40 | 0.3 | 48 | 56 500 | 102 000 | 5 750 | 10 400 | 6 500 | 10 000 | **RNAO-40×55×40ZW** |
| **45** | +0.041 +0.025 | 55 | 17 | 0.3 | 50 | 22 300 | 44 500 | 2 280 | 4 550 | 6 000 | 9 000 | **RNAO-45×55×17** |
| | | 55 | 34 | 0.3 | 50 | 38 500 | 89 500 | 3 900 | 9 100 | 6 000 | 9 000 | **RNAO-45×55×34ZW** |
| | | 62 | 20 | 0.3 | 53 | 36 000 | 59 000 | 3 650 | 6 000 | 6 000 | 9 000 | **RNAO-45×62×20** |
| | | 62 | 40 | 0.3 | 53 | 61 500 | 118 000 | 6 250 | 12 000 | 6 000 | 9 000 | **RNAO-45×62×40ZW** |
| **50** | +0.041 +0.025 | 62 | 20 | 0.3 | 55 | 27 900 | 62 000 | 2 850 | 6 300 | 5 500 | 8 000 | **RNAO-50×62×20** |
| | | 62 | 40 | 0.3 | 55 | 48 000 | 124 000 | 4 900 | 12 600 | 5 500 | 8 000 | **RNAO-50×62×40ZW** |
| | | 65 | 20 | 0.3 | 58 | 38 500 | 67 500 | 3 950 | 6 850 | 5 500 | 8 000 | **RNAO-50×65×20** |
| | | 65 | 40 | 0.6 | 58 | 66 500 | 135 000 | 6 750 | 13 700 | 5 500 | 8 000 | **RNAO-50×65×40ZW** |
| **55** | +0.049 +0.030 | 68 | 20 | 0.6 | 60 | 28 800 | 66 500 | 2 940 | 6 750 | 4 800 | 7 500 | **RNAO-55×68×20** |
| | | 68 | 25 | 0.6 | 63 | 50 500 | 97 500 | 5 150 | 9 950 | 4 800 | 7 500 | **RNAO-55×68×25** |
| | | 68 | 40 | 0.6 | 60 | 49 500 | 133 000 | 5 050 | 13 500 | 4 800 | 7 500 | **RNAO-55×68×40ZW** |
| | | 72 | 20 | 0.6 | 63 | 39 000 | 70 000 | 3 950 | 7 100 | 4 800 | 7 500 | **RNAO-55×72×20** |
| | | 72 | 40 | 0.6 | 63 | 66 500 | 140 000 | 6 800 | 14 200 | 4 800 | 7 500 | **RNAO-55×72×40ZW** |
| **60** | +0.049 +0.030 | 75 | 46 | 1 | 68 | 76 000 | 170 000 | 7 750 | 17 400 | 4 400 | 6 500 | **RNAO-60×75×46ZW** |
| | | 78 | 20 | 1 | 68 | 40 000 | 75 000 | 4 100 | 7 650 | 4 400 | 6 500 | **RNAO-60×78×20** |
| | | 78 | 40 | 1 | 68 | 69 000 | 150 000 | 7 050 | 15 300 | 4 400 | 6 500 | **RNAO-60×78×40ZW** |
| **65** | +0.049 +0.030 | 85 | 30 | 1 | 73 | 61 000 | 132 000 | 6 200 | 13 400 | 4 100 | 6 000 | **RNAO-65×85×30** |
| | | 85 | 60 | 1 | 73 | 104 000 | 263 000 | 10 600 | 26 800 | 4 100 | 6 000 | **RNAO-65×85×60ZW** |
| **70** | +0.049 +0.030 | 90 | 30 | 1 | 78 | 65 500 | 149 000 | 6 700 | 15 200 | 3 800 | 5 500 | **RNAO-70×90×30** |
| | | 90 | 60 | 1 | 78 | 112 000 | 297 000 | 11 500 | 30 500 | 3 800 | 5 500 | **RNAO-70×90×60ZW** |
| **75** | +0.049 +0.030 | 95 | 30 | 1 | 83 | 67 500 | 157 000 | 6 850 | 16 100 | 3 600 | 5 500 | **RNAO-75×95×30** |
| | | 95 | 60 | 1 | 83 | 115 000 | 315 000 | 11 800 | 32 000 | 3 600 | 5 500 | **RNAO-75×95×60ZW** |
| **80** | +0.049 +0.030 | 95 | 30 | 1 | 86 | 57 000 | 159 000 | 5 800 | 16 200 | 3 300 | 5 000 | **RNAO-80×95×30** |
| | | 95 | 56 | 1 | 88 | 105 000 | 284 000 | 10 700 | 29 000 | 3 300 | 5 000 | **RNAO-80×95×56ZW** |
| | | 100 | 30 | 1 | 88 | 69 000 | 166 000 | 7 050 | 17 000 | 3 300 | 5 000 | **RNAO-80×100×30** |
| | | 100 | 60 | 1 | 88 | 119 000 | 335 000 | 12 100 | 34 000 | 3 300 | 5 000 | **RNAO-80×100×60ZW** |
| **85** | +0.058 +0.036 | 105 | 25 | 1 | 93 | 61 500 | 146 000 | 6 250 | 14 900 | 3 100 | 4 700 | **RNAO-85×105×25** |

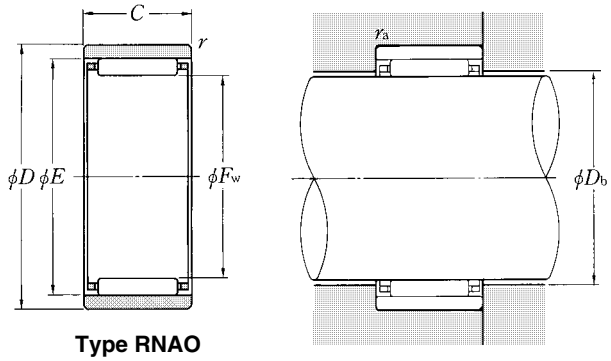
Note 1) Allowable minimum chamfer dimension $r$.




| Abutment dimensions mm | | | | Mass kg |
|---|---|---|---|---|
| $d_b$ | $D_a$ max | $D_b$ | $r_{as}$ max | (approx.) |
| 47.2 | 53 | 40.6 | 0.3 | 0.275 |
| 49.2 | 53 | 45.6 | 0.3 | 0.083 |
| 49.2 | 53 | 45.6 | 0.3 | 0.165 |
| 52.2 | 60 | 45.6 | 0.3 | 0.175 |
| 52.2 | 60 | 45.6 | 0.3 | 0.377 |
| 54.2 | 60 | 50.6 | 0.3 | 0.140 |
| 54.2 | 60 | 50.6 | 0.3 | 0.295 |
| 57.2 | 63 | 50.6 | 0.3 | 0.168 |
| 57.2 | 61 | 50.6 | 0.6 | 0.355 |
| 59.4 | 64 | 55.8 | 0.6 | 0.166 |
| 62.4 | 64 | 55.8 | 0.6 | 0.200 |
| 59.4 | 64 | 55.8 | 0.6 | 0.310 |
| 62.4 | 68 | 55.8 | 0.6 | 0.216 |
| 62.4 | 68 | 55.8 | 0.6 | 0.425 |
| 67.2 | 70 | 60.8 | 1 | 0.461 |
| 67.2 | 73 | 60.8 | 1 | 0.255 |
| 67.2 | 73 | 60.8 | 1 | 0.500 |
| 72.2 | 80 | 66 | 1 | 0.464 |
| 72.2 | 80 | 66 | 1 | 0.951 |
| 77.2 | 85 | 71 | 1 | 0.499 |
| 77.2 | 85 | 71 | 1 | 1.00 |
| 82.2 | 90 | 76 | 1 | 0.520 |
| 82.2 | 90 | 76 | 1 | 1.04 |
| 85.2 | 90 | 81 | 1 | 0.405 |
| 87.2 | 90 | 81 | 1 | 0.755 |
| 87.2 | 95 | 81 | 1 | 0.580 |
| 87.2 | 95 | 81 | 1 | 1.10 |
| 92.2 | 100 | 86 | 1 | 0.459 |

## Without inner ring

### Type RNAO

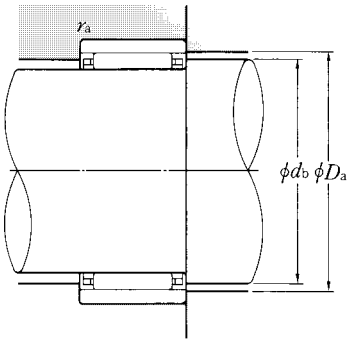



**Type RNAO**

$F_w$ 85～100mm

| Boundary dimensions mm | | | | | | Basic load ratings N dynamic | | Basic load ratings kgf dynamic | | Limiting speeds r/min grease | Limiting speeds r/min oil | Bearing numbers |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $F_w$ | | $D$ | $C$ | $r_{s\,min}$ [1] | $E$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | |
| **85** | $^{+0.058}_{+0.036}$ | 105 | 30 | 1 | 93 | 71 000 | 175 000 | 7 200 | 17 900 | 3 100 | 4 700 | **RNAO- 85×105×30** |
| **90** | $^{+0.058}_{+0.036}$ | 105 | 26 | 1 | 98 | 64 000 | 157 000 | 6 550 | 16 000 | 3 000 | 4 400 | **RNAO- 90×105×26** |
| | | 110 | 30 | 1 | 98 | 72 500 | 184 000 | 7 400 | 18 800 | 3 000 | 4 400 | **RNAO- 90×110×30** |
| **95** | $^{+0.058}_{+0.036}$ | 115 | 30 | 1 | 103 | 74 000 | 193 000 | 7 550 | 19 600 | 2 800 | 4 200 | **RNAO- 95×115×30** |
| **100** | $^{+0.058}_{+0.035}$ | 120 | 30 | 1 | 108 | 76 000 | 201 000 | 7 700 | 20 500 | 2 700 | 4 000 | **RNAO-100×120×30** |

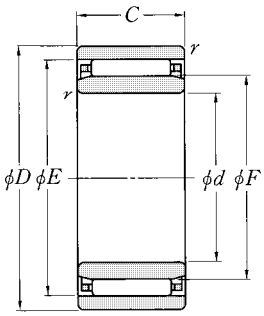
Note 1) Allowable minimum chamfer dimension $r$.



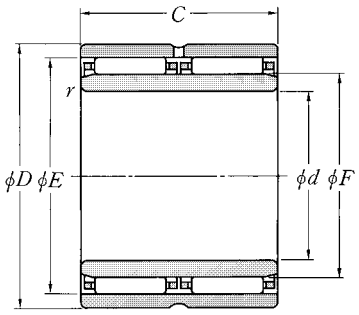

| Abutment dimensions mm | | | | Mass kg |
|---|---|---|---|---|
| $d_b$ | $D_a$ max | $D_b$ | $r_{as}$ max | (approx.) |
| 92.2 | 100 | 86 | 1 | 0.585 |
| 97.2 | 100 | 91 | 1 | 0.373 |
| 97.2 | 105 | 91 | 1 | 0.610 |
| 102.2 | 110 | 96 | 1 | 0.640 |
| 107.2 | 115 | 101 | 1 | 0.694 |

## With inner ring

### Type NAO
### Type NAO··ZW




Type NAO

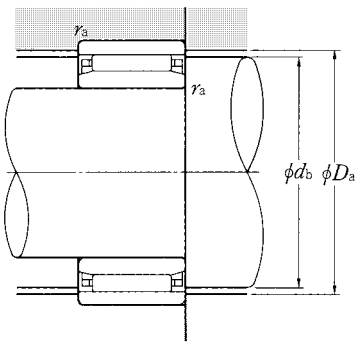


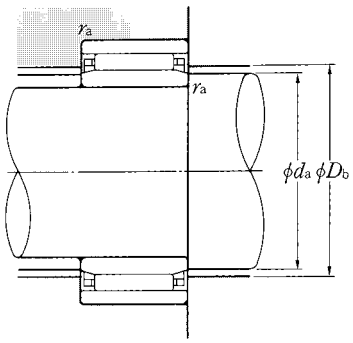

Type NAO··ZW

$d$ 6～30mm

| Boundary dimensions mm | | | | | | Basic load ratings N dynamic | | kgf dynamic | | Limiting speeds r/min grease | oil | Bearing numbers |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $d$ | $D$ | $C$ | $r_{s\,min}$ [1)] | $F$ | $E$ | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | |
| **6** | 17 | 10 | 0.3 | 10 | 13 | 4 550 | 5 100 | 460 | 520 | 19 000 | 28 000 | **NAO- 6×17×10T2** |
| **7** | 20 | 12 | 0.3 | 10 | 16 | 7 100 | 5 950 | 720 | 610 | 19 000 | 28 000 | **NAO- 7×20×12** |
| **9** | 22 | 12 | 0.3 | 12 | 18 | 8 650 | 8 000 | 880 | 815 | 17 000 | 26 000 | **NAO- 9×22×12** |
| **10** | 22 | 13 | 0.3 | 14 | 18 | 8 300 | 10 100 | 845 | 1 030 | 16 000 | 24 000 | **NAO-10×22×13** |
| | 22 | 20 | 0.3 | 14 | 18 | 11 800 | 16 000 | 1 210 | 1 630 | 16 000 | 24 000 | **NAO-10×22×20ZW** |
| | 26 | 12 | 0.3 | 14 | 20 | 9 350 | 9 150 | 955 | 930 | 16 000 | 24 000 | **NAO-10×26×12** |
| **12** | 24 | 13 | 0.3 | 16 | 20 | 9 050 | 11 800 | 925 | 1 200 | 15 000 | 23 000 | **NAO-12×24×13** |
| | 24 | 20 | 0.3 | 16 | 20 | 12 900 | 18 500 | 1 310 | 1 890 | 15 000 | 23 000 | **NAO-12×24×20ZW** |
| | 28 | 12 | 0.3 | 16 | 22 | 11 700 | 12 500 | 1 190 | 1 280 | 15 000 | 23 000 | **NAO-12×28×12** |
| **15** | 28 | 13 | 0.3 | 20 | 24 | 10 000 | 14 300 | 1 020 | 1 460 | 13 000 | 20 000 | **NAO-15×28×13** |
| | 28 | 26 | 0.3 | 20 | 24 | 17 100 | 28 600 | 1 750 | 2 910 | 13 000 | 20 000 | **NAO-15×28×26ZW** |
| | 32 | 12 | 0.3 | 20 | 26 | 12 900 | 15 100 | 1 320 | 1 540 | 13 000 | 20 000 | **NAO-15×32×12** |
| **17** | 30 | 13 | 0.3 | 22 | 26 | 10 200 | 15 200 | 1 040 | 1 550 | 12 000 | 18 000 | **NAO-17×30×13** |
| | 30 | 26 | 0.3 | 22 | 26 | 17 500 | 30 500 | 1 790 | 3 100 | 12 000 | 18 000 | **NAO-17×30×26ZW** |
| | 35 | 16 | 0.3 | 22 | 29 | 18 700 | 22 700 | 1 910 | 2 310 | 12 000 | 18 000 | **NAO-17×35×16** |
| | 35 | 32 | 0.3 | 22 | 29 | 32 000 | 45 500 | 3 300 | 4 650 | 12 000 | 18 000 | **NAO-17×35×32ZW** |
| **20** | 35 | 17 | 0.3 | 25 | 29 | 14 200 | 24 000 | 1 450 | 2 450 | 11 000 | 16 000 | **NAO-20×35×17** |
| | 35 | 26 | 0.3 | 25 | 29 | 18 400 | 33 500 | 1 880 | 3 450 | 11 000 | 16 000 | **NAO-20×35×26ZW** |
| | 37 | 16 | 0.3 | 25 | 32 | 19 500 | 24 700 | 1 990 | 2 520 | 11 000 | 16 000 | **NAO-20×37×16** |
| | 37 | 32 | 0.3 | 25 | 32 | 33 500 | 49 500 | 3 400 | 5 050 | 11 000 | 16 000 | **NAO-20×37×32ZW** |
| **25** | 40 | 17 | 0.3 | 30 | 35 | 19 400 | 32 500 | 1 970 | 3 350 | 9 000 | 13 000 | **NAO-25×40×17** |
| | 40 | 26 | 0.3 | 30 | 35 | 25 200 | 46 000 | 2 570 | 4 650 | 9 000 | 13 000 | **NAO-25×40×26ZW** |
| | 42 | 16 | 0.3 | 30 | 37 | 21 900 | 30 500 | 2 230 | 3 100 | 9 000 | 13 000 | **NAO-25×42×16** |
| | 42 | 32 | 0.3 | 30 | 37 | 37 500 | 60 500 | 3 850 | 6 200 | 9 000 | 13 000 | **NAO-25×42×32ZW** |
| **29** | 42 | 13 | 0.3 | 32 | 37 | 14 500 | 23 000 | 1 480 | 2 350 | 8 500 | 13 000 | **NAO-29×42×13** |
| **30** | 45 | 13 | 0.3 | 35 | 40 | 15 200 | 25 100 | 1 550 | 2 560 | 7 500 | 11 000 | **NAO-30×45×13** |
| | 45 | 17 | 0.3 | 35 | 40 | 20 000 | 36 000 | 2 040 | 3 650 | 7 500 | 11 000 | **NAO-30×45×17** |
| | 45 | 26 | 0.3 | 35 | 40 | 26 100 | 50 000 | 2 660 | 5 100 | 7 500 | 11 000 | **NAO-30×45×26ZW** |

Note 1) Allowable minimum chamfer dimension $r$.

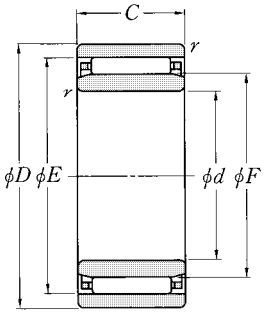


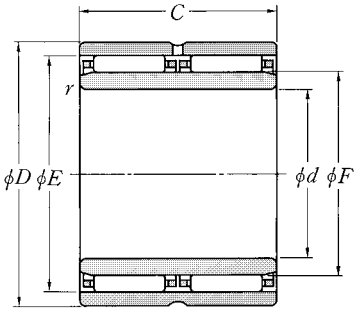

| Abutment dimensions mm | | | | | Mass kg |
|---|---|---|---|---|---|
| $d_a$ min | $d_b$ | $D_a$ max | $D_b$ | $r_{as}$ max | (approx.) |
| 8 | 12.7 | 15 | 10.3 | 0.3 | 0.014 |
| 9 | 15.7 | 18 | 10.3 | 0.3 | 0.022 |
| 11 | 17.6 | 20 | 12.3 | 0.3 | 0.024 |
| 12 | 17.6 | 20 | 14.4 | 0.3 | 0.026 |
| 12 | 17.6 | 20 | 14.4 | 0.3 | 0.039 |
| 12 | 19.6 | 24 | 14.4 | 0.3 | 0.036 |
| 14 | 19.6 | 22 | 16.4 | 0.3 | 0.030 |
| 14 | 19.6 | 22 | 16.4 | 0.3 | 0.044 |
| 14 | 21.6 | 26 | 16.4 | 0.3 | 0.040 |
| 17 | 23.6 | 26 | 20.4 | 0.3 | 0.029 |
| 17 | 23.6 | 26 | 20.4 | 0.3 | 0.075 |
| 17 | 25.6 | 30 | 20.4 | 0.3 | 0.050 |
| 19 | 25.6 | 28 | 22.4 | 0.3 | 0.042 |
| 19 | 25.6 | 28 | 22.4 | 0.3 | 0.081 |
| 19 | 28.4 | 33 | 22.4 | 0.3 | 0.078 |
| 19 | 28.4 | 33 | 22.4 | 0.3 | 0.148 |
| 22 | 28.4 | 33 | 25.6 | 0.3 | 0.076 |
| 22 | 28.4 | 33 | 25.6 | 0.3 | 0.112 |
| 22 | 31.4 | 35 | 25.6 | 0.3 | 0.082 |
| 22 | 31.4 | 35 | 25.6 | 0.3 | 0.155 |
| 27 | 34.4 | 38 | 30.6 | 0.3 | 0.088 |
| 27 | 34.4 | 38 | 30.6 | 0.3 | 0.130 |
| 27 | 36.4 | 40 | 30.6 | 0.3 | 0.086 |
| 27 | 36.4 | 40 | 30.6 | 0.3 | 0.190 |
| 31 | 36.4 | 40 | 32.6 | 0.3 | 0.062 |
| 32 | 39.4 | 43 | 35.6 | 0.3 | 0.077 |
| 32 | 39.4 | 43 | 35.6 | 0.3 | 0.102 |
| 32 | 39.4 | 43 | 35.6 | 0.3 | 0.157 |

## With inner ring

### Type NAO
### Type NAO··ZW




Type NAO

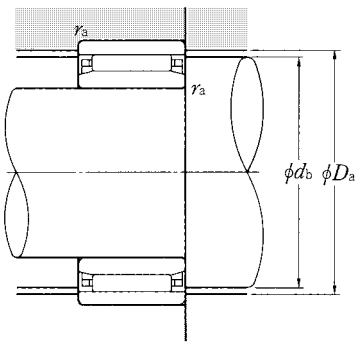


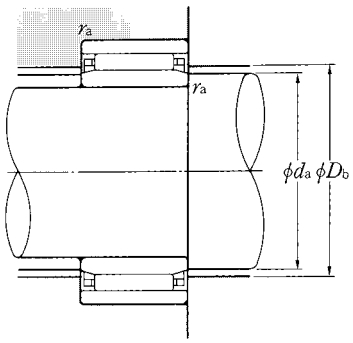

Type NAO··ZW

*d* 30～65mm

| Boundary dimensions mm | | | | | | Basic load ratings N dynamic | | kgf dynamic | | Limiting speeds r/min grease | oil | Bearing numbers |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *d* | *D* | *C* | $r_{s\,min}$ [1] | *F* | *E* | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | |
| 30 | 47 | 16 | 0.3 | 35 | 42 | 24 100 | 36 000 | 2 450 | 3 650 | 7 500 | 11 000 | NAO-30×47×16 |
| | 47 | 18 | 0.3 | 35 | 42 | 24 700 | 37 000 | 2 510 | 3 750 | 7 500 | 11 000 | NAO-30×47×18 |
| | 47 | 32 | 0.3 | 35 | 42 | 41 500 | 71 500 | 4 200 | 3 700 | 7 500 | 11 000 | NAO-30×47×32ZW |
| | 52 | 18 | 0.3 | 37 | 44 | 26 300 | 41 000 | 2 680 | 4 150 | 7 000 | 11 000 | NAO-30×52×18 |
| 33 | 47 | 13 | 0.3 | 37 | 42 | 15 900 | 27 100 | 1 620 | 2 770 | 7 000 | 11 000 | NAO-33×47×13 |
| 35 | 50 | 17 | 0.3 | 40 | 45 | 21 800 | 41 500 | 2 220 | 4 250 | 6 500 | 10 000 | NAO-35×50×17 |
| | 50 | 34 | 0.3 | 40 | 45 | 37 500 | 83 000 | 3 800 | 8 500 | 6 500 | 10 000 | NAO-35×50×34ZW |
| | 55 | 20 | 0.3 | 40 | 47 | 31 000 | 51 500 | 3 150 | 5 250 | 6 500 | 10 000 | NAO-35×55×20 |
| | 55 | 40 | 0.3 | 40 | 48 | 56 500 | 102 000 | 5 750 | 10 400 | 6 500 | 10 000 | NAO-35×55×40ZW |
| 40 | 55 | 17 | 0.3 | 45 | 50 | 22 300 | 44 500 | 2 280 | 4 550 | 6 000 | 9 000 | NAO-40×55×17 |
| | 55 | 34 | 0.3 | 45 | 50 | 38 500 | 89 500 | 3 900 | 9 100 | 6 000 | 9 000 | NAO-40×55×34ZW |
| | 62 | 20 | 0.3 | 45 | 53 | 36 000 | 59 000 | 3 650 | 6 000 | 6 000 | 9 000 | NAO-40×62×20 |
| | 62 | 40 | 0.3 | 45 | 53 | 61 500 | 118 000 | 6 250 | 12 000 | 6 000 | 9 000 | NAO-40×62×40ZW |
| | 65 | 20 | 0.3 | 50 | 58 | 38 500 | 67 500 | 3 950 | 6 850 | 5 500 | 8 000 | NAO-40×65×20 |
| 45 | 62 | 20 | 0.3 | 50 | 55 | 27 900 | 62 000 | 2 850 | 6 300 | 5 500 | 8 000 | NAO-45×62×20 |
| | 62 | 40 | 0.3 | 50 | 55 | 48 000 | 124 000 | 4 900 | 12 600 | 5 500 | 8 000 | NAO-45×62×40ZW |
| | 72 | 20 | 0.6 | 55 | 63 | 39 000 | 70 000 | 3 950 | 7 100 | 4 800 | 7 500 | NAO-45×72×20 |
| | 72 | 40 | 0.6 | 55 | 63 | 66 500 | 140 000 | 6 800 | 14 200 | 4 800 | 7 500 | NAO-45×72×40ZW |
| 50 | 68 | 20 | 0.6 | 55 | 60 | 28 800 | 66 500 | 2 940 | 6 750 | 4 800 | 7 500 | NAO-50×68×20 |
| | 68 | 40 | 0.6 | 55 | 60 | 49 500 | 133 000 | 5 050 | 13 500 | 4 800 | 7 500 | NAO-50×68×40ZW |
| | 78 | 20 | 1 | 60 | 68 | 40 000 | 75 000 | 4 100 | 7 650 | 4 400 | 6 500 | NAO-50×78×20 |
| | 78 | 40 | 1 | 60 | 68 | 69 000 | 150 000 | 7 050 | 15 300 | 4 400 | 6 500 | NAO-50×78×40ZW |
| 55 | 85 | 30 | 1 | 65 | 73 | 61 000 | 132 000 | 6 200 | 13 400 | 4 100 | 6 000 | NAO-55×85×30 |
| | 85 | 60 | 1 | 65 | 73 | 104 000 | 263 000 | 10 600 | 26 800 | 4 100 | 6 000 | NAO-55×85×60ZW |
| 60 | 90 | 30 | 1 | 70 | 78 | 65 500 | 149 000 | 6 700 | 15 200 | 3 800 | 5 500 | NAO-60×90×30 |
| | 90 | 60 | 1 | 70 | 78 | 112 000 | 297 000 | 11 500 | 30 500 | 3 800 | 5 500 | NAO-60×90×60ZW |
| 65 | 95 | 30 | 1 | 75 | 83 | 67 500 | 157 000 | 6 850 | 16 100 | 3 600 | 5 500 | NAO-65×95×30 |
| | 95 | 60 | 1 | 75 | 83 | 115 000 | 315 000 | 11 800 | 32 000 | 3 600 | 5 500 | NAO-65×95×60ZW |

Note 1) Allowable minimum chamfer dimension $r$.

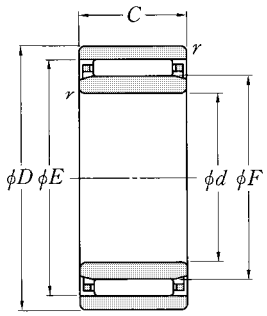


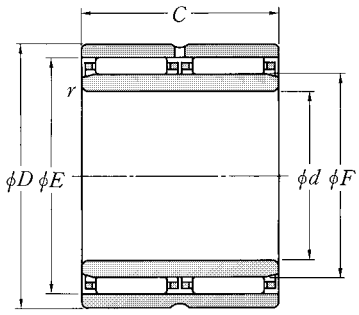

| Abutment dimensions mm | | | | | Mass kg |
|---|---|---|---|---|---|
| $d_a$ min | $d_b$ | $D_a$ max | $D_b$ | $r_{as}$ max | (approx.) |
| 32 | 41.4 | 45 | 35.6 | 0.3 | 0.109 |
| 32 | 41.4 | 45 | 35.6 | 0.3 | 0.119 |
| 32 | 41.4 | 45 | 35.6 | 0.3 | 0.205 |
| 32 | 43.4 | 50 | 37.6 | 0.3 | 0.177 |
| 35 | 41.4 | 45 | 37.6 | 0.3 | 0.085 |
| 37 | 44.4 | 48 | 40.6 | 0.3 | 0.113 |
| 37 | 44.4 | 48 | 40.6 | 0.3 | 0.225 |
| 37 | 46.2 | 53 | 40.6 | 0.3 | 0.190 |
| 37 | 47.2 | 53 | 40.6 | 0.3 | 0.360 |
| 42 | 49.2 | 53 | 45.6 | 0.3 | 0.127 |
| 42 | 49.2 | 53 | 45.6 | 0.3 | 0.250 |
| 42 | 52.2 | 60 | 45.6 | 0.3 | 0.230 |
| 42 | 52.2 | 60 | 45.6 | 0.3 | 0.385 |
| 42 | 57.2 | 63 | 50.6 | 0.3 | 0.279 |
| 47 | 54.2 | 60 | 50.6 | 0.3 | 0.192 |
| 47 | 54.2 | 60 | 50.6 | 0.3 | 0.385 |
| 49 | 62.4 | 68 | 55.8 | 0.6 | 0.335 |
| 49 | 62.4 | 68 | 55.8 | 0.6 | 0.660 |
| 54 | 59.4 | 64 | 55.8 | 0.6 | 0.230 |
| 54 | 59.4 | 64 | 55.8 | 0.6 | 0.440 |
| 55 | 67.2 | 73 | 60.8 | 1 | 0.410 |
| 55 | 67.2 | 73 | 60.8 | 1 | 0.755 |
| 60 | 72.2 | 80 | 66 | 1 | 0.680 |
| 60 | 72.2 | 80 | 66 | 1 | 1.35 |
| 65 | 77.2 | 85 | 71 | 1 | 0.720 |
| 65 | 77.2 | 85 | 71 | 1 | 1.45 |
| 70 | 82.2 | 90 | 76 | 1 | 0.770 |
| 70 | 82.2 | 90 | 76 | 1 | 1.54 |

## With inner ring

### Type NAO
### Type NAO··ZW




Type NAO

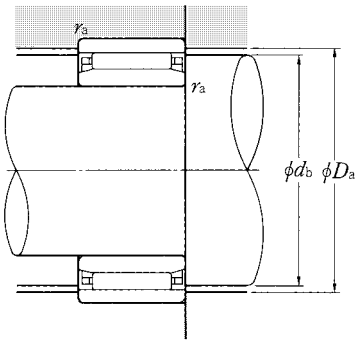


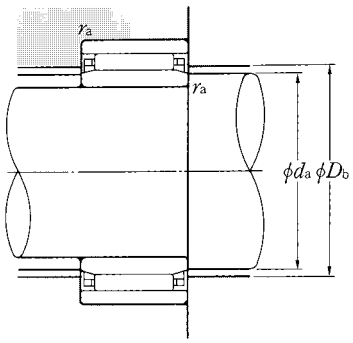

Type NAO··ZW

*d* 70～90mm

| Boundary dimensions mm | | | | | | Basic load ratings dynamic N | static N | dynamic kgf | static kgf | Limiting speeds r/min grease | Limiting speeds r/min oil | Bearing numbers |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *d* | *D* | *C* | $r_{s\,min}$ [1] | *F* | *E* | $C_r$ | $C_{or}$ | $C_r$ | $C_{or}$ | | | |
| 70 | 95 | 30 | 1 | 80 | 86 | 57 000 | 159 000 | 5 800 | 16 200 | 3 300 | 5 000 | NAO-70×95×30 |
| | 95 | 56 | 1 | 80 | 88 | 105 000 | 284 000 | 10 700 | 29 000 | 3 300 | 5 000 | NAO-70×95×56ZW |
| | 100 | 30 | 1 | 80 | 88 | 69 000 | 166 000 | 7 050 | 17 000 | 3 300 | 5 000 | NAO-70×100×30 |
| | 100 | 60 | 1 | 80 | 88 | 119 000 | 335 000 | 12 100 | 34 000 | 3 300 | 5 000 | NAO-70×100×60ZW |
| 75 | 105 | 25 | 1 | 85 | 93 | 61 500 | 146 000 | 6 250 | 14 900 | 3 100 | 4 700 | NAO-75×105×25 |
| | 105 | 30 | 1 | 85 | 93 | 71 000 | 175 000 | 7 200 | 17 900 | 3 100 | 4 700 | NAO-75×105×30 |
| 80 | 110 | 30 | 1 | 90 | 98 | 72 500 | 184 000 | 7 400 | 18 800 | 3 000 | 4 400 | NAO-80×110×30 |
| 85 | 115 | 30 | 1 | 95 | 103 | 74 000 | 193 000 | 7 550 | 19 600 | 2 800 | 4 200 | NAO-85×115×30 |
| 90 | 120 | 30 | 1 | 100 | 108 | 76 000 | 201 000 | 7 700 | 20 500 | 2 700 | 4 000 | NAO-90×120×30 |

Note 1) Allowable minimum chamfer dimension $r$.




| Abutment dimensions mm | | | | | Mass kg |
|---|---|---|---|---|---|
| $d_a$ min | $d_b$ | $D_a$ max | $D_b$ | $r_{as}$ max | (approx.) |
| 75 | 85.2 | 90 | 81 | 1 | 0.675 |
| 75 | 87.2 | 90 | 81 | 1 | 1.26 |
| 75 | 87.2 | 95 | 81 | 1 | 0.850 |
| 75 | 87.2 | 95 | 81 | 1 | 1.70 |
| 80 | 92.2 | 100 | 86 | 1 | 0.700 |
| 80 | 92.2 | 100 | 86 | 1 | 0.880 |
| 85 | 97.2 | 105 | 91 | 1 | 0.920 |
| 90 | 102.2 | 110 | 96 | 1 | 0.960 |
| 95 | 107.2 | 115 | 101 | 1 | 1.04 |